

Lab 2 Exercise Solutions

SOCIOL 723, Social Statistics II

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```
library(dplyr); library(ggplot2); library(sandwich)
gss <- readRDS("../Data/gss_earnings.rds")
m <- lm(lnearn ~ educ + exper + I(exper^2) + female + black, data = gss)
```

Exercise 1: Coverage of the classical interval

Same design as the lab, with the heteroskedasticity turned up from $0.5 \exp(0.8x)$ to $0.5 \exp(1.5x)$.

```
set.seed(723)
sim_once <- function(n = 200, expo = 1.5) {
  x <- rnorm(n)
  e <- rnorm(n, sd = 0.5 * exp(expo * x))
  mm <- lm(I(1 + 0.5 * x + e) ~ x)
  c(b = coef(mm)[2],
    sc = sqrt(diag(vcov(mm)))[2],
    s3 = sqrt(diag(vcovHC(mm, type = "HC3")))[2])
}
reps <- t(replicate(2000, sim_once()))
covers <- function(b, se) mean(abs(b - 0.5) / se < 1.96)
c(classical = covers(reps[, 1], reps[, 2]),
  hc3       = covers(reps[, 1], reps[, 3]))
#> classical      hc3
#>      0.556      0.959
```

Classical coverage collapses far below its nominal 95%: the error variance now explodes where x is large, exactly where the meat's weights d_i^2 are biggest, so the pooled s^2 understates the variance badly. HC3 loses some ground too — with heteroskedasticity this violent, the meat is estimated from a handful of extreme observations — but it stays far closer to nominal. The lesson survives: the classical interval's failure is structural, the robust interval's is finite-sample.

Exercise 2: HC3 versus HC0 in small samples

```
set.seed(723)
one_n <- function(n) {
  reps <- t(replicate(2000, {
    x <- rnorm(n)
    e <- rnorm(n, sd = 0.5 * exp(0.8 * x))
    mm <- lm(I(1 + 0.5 * x + e) ~ x)
    c(b = coef(mm)[2],
      s0 = sqrt(diag(vcovHC(mm, type = "HC0")))[2],
      s3 = sqrt(diag(vcovHC(mm, type = "HC3")))[2])
  }))
  c(n = n, hc0 = covers(reps[, 1], reps[, 2]),
    hc3 = covers(reps[, 1], reps[, 3]))
}
round(t(sapply(c(30, 60, 120, 500), one_n)), 3)
#>      n   hc0   hc3
#> [1,]  30 0.888 0.935
#> [2,]  60 0.922 0.944
#> [3,] 120 0.929 0.940
#> [4,] 500 0.950 0.953
```

At $n = 30$ HC0 under-covers noticeably while HC3's leverage rescaling buys several points of coverage; by $n = 120$ the gap has nearly closed, and at $n = 500$ the two are indistinguishable. That is the general pattern behind the lecture's advice: the HC variants are a small-sample story, and all of them coincide as n grows.

Exercise 3: Choosing the cluster level

A state-level minimum-wage law is one treatment draw shared by everyone in the state, so the lottery runs over states: **cluster by state**. This is coarser than the GSS's own sampling clusters (PSUs sit inside states), and with nested lotteries the coarsest one covers both — same-PSU pairs are also same-state pairs, so state clustering keeps the sampling covariances too.

If the law varies by state *and* year, the tempting answer — cluster on state-year cells — is too fine: a state's errors are strongly correlated *across* years (the Bertrand-Duflo-Mullainathan lesson), and state-year clustering assumes those covariances away. The safe answer remains clustering by state, which allows arbitrary correlation over time within each state.

Exercise 4: A statistic with no easy formula

```
set.seed(723)
turning <- function(fit) -coef(fit)["exper"] / (2 * coef(fit)["I(exper^2)"])
```

```

turning(m)
#> exper
#> 35.77

B <- 5000
tp_b <- replicate(B, {
  idx <- sample(nrow(gss), replace = TRUE)
  turning(lm(lnearn ~ educ + exper + I(exper^2) + female + black,
    data = gss[idx, ]))
})

c(estimate = turning(m), se_boot = sd(tp_b))
#> estimate.exper      se_boot
#>      35.773      2.301
c(normal_approx = turning(m) + c(-1.96, 1.96) * sd(tp_b))
#> normal_approx1 normal_approx2
#>      31.26      40.28
quantile(tp_b, c(.025, .975))
#> 2.5% 97.5%
#> 32.52 41.56

```

```

tibble::tibble(tp = tp_b) |>
  ggplot(aes(tp)) +
  geom_histogram(bins = 60, fill = "navy", alpha = .65) +
  geom_vline(xintercept = turning(m), colour = "firebrick") +
  labs(x = "bootstrap replicate of the turning point (years)", y = "count") +
  theme_minimal(base_size = 10)

```

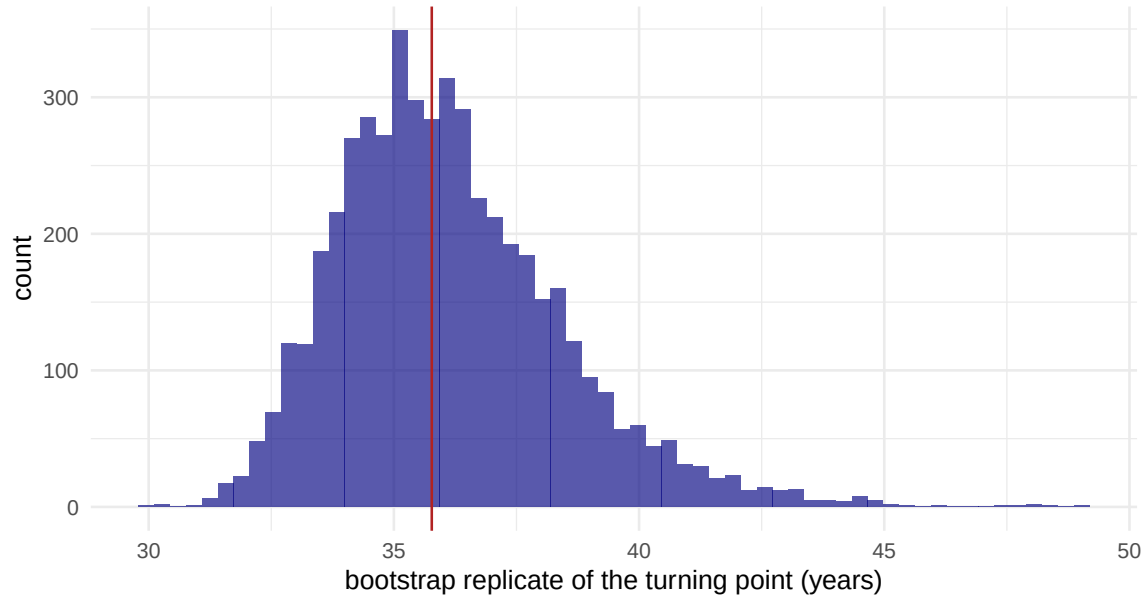


Figure 1: Bootstrap distribution of the experience-profile turning point, $B = 5,000$.

The turning point lands around 36 years of experience. The histogram is close to bell-shaped — the denominator $2\hat{\beta}_{\text{exper}^2}$ is many standard errors from zero, so the ratio behaves well — and accordingly the normal-approximation and percentile intervals nearly agree. When they agree, either is defensible; the percentile interval is still the safer habit, because it would have followed the skew had there been any.